

DATA WAREHOUSE PROJECT :

Advertising analyse

Technical Project Report - DW

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1. Description of the Domain and Process

The domain of this project is Digital Marketing Analytics. In the current digital economy, companies deploy advertising content across multiple platforms (such as Facebook and Instagram) to reach specific demographics. The objective of this project is to build a data warehouse, analyze the collected data, and create visualizations to support recommendations for future advertising campaigns.

1.1 Process Covered

The Data Warehouse covers the end-to-end lifecycle of digital advertisements:

- **Campaign Planning:** Defining budgets, durations, and seasonal themes (e.g., "Summer Launch" vs. "Winter Promo").
- **Ad Placement:** Distributing specific creative assets (Videos, Stories, Carousels) targeted at specific audience segments (by age, gender, and interests).
- **User Engagement:** Tracking real-time interactions from users, ranging from passive views (Impressions) to active engagement (Clicks, Likes, Shares) and final conversions (Purchases).
- **Audience Demographics:** Mapping these interactions back to user profiles (country, age group, specific interests) to see who is actually responding to the ads.

1.2 Motivation for Building the Data Warehouse (DW)

The specific motivations are:

- **ROI Analysis:** Calculating the Return on Ad Spend (ROAS) by comparing campaign budgets against the resulting purchase events.
- **Temporal Trends:** Analyzing how ad performance fluctuates by time of day or day of the week to optimize scheduling

2. Description of Used Technologies

The main constraint I have is the operating system, I use Linux (Fedora 42), and the majority of tools use for data warehouse and BI like Microsoft Access, Tableau are not available on this system.

So an open-source stack has been selected:

2.1 Database & Storage

- SQLite: It is native to Linux, requires no server setup (serverless), and stores the entire database in a single portable file that I can easily use on my visualization app

2.2 Business Intelligence (BI) & Visualization

- Apache Superset: A high-performance visualization engine.

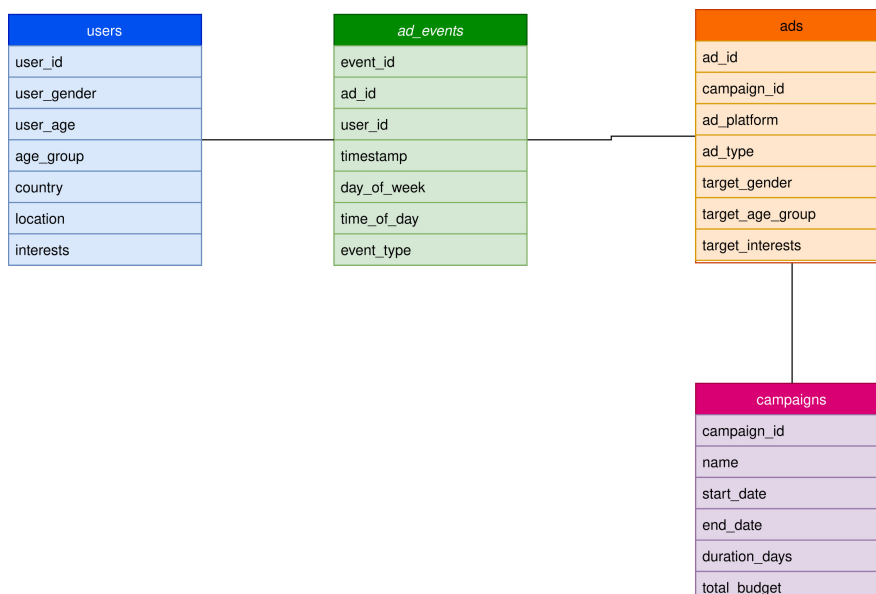
2.3 Data Processing (ETL)

Python: Using libraries like pandas or sqlite3 on Linux to automate the cleaning, joining, and loading of the CSV files into the SQLite Star Schema.

3. Description of the dataset

The data warehouse relies on a single relational data source, initially stored and managed within a SQLite database. This source dataset captures marketing interactions and consists of four distinct but interrelated tables that track users, advertisements, campaigns, and user actions.

- **users:** Contains demographic details of the platform's audience, including unique identifiers (`user_id`), age, gender, geographic location (country, location), and segmented user interests.
- **ad_events:** The central transactional table recording every interaction. It tracks an `event_id`, the associated `user_id`, the `ad_id` involved, a precise timestamp, and the `event_type` (e.g., impressions, clicks).
- **ads:** Stores structural configurations for specific advertisements, detailing the platform used (`ad_platform`), the format (`ad_type`), and target criteria such as `target_gender`, `target_age_group`, and `target_interests`.
- **campaigns:** Holds macro-level marketing metadata linked to ads via `campaign_id`, mapping out financial parameters (`total_budget`), operational windows (`start_date`, `end_date`), and overall duration.

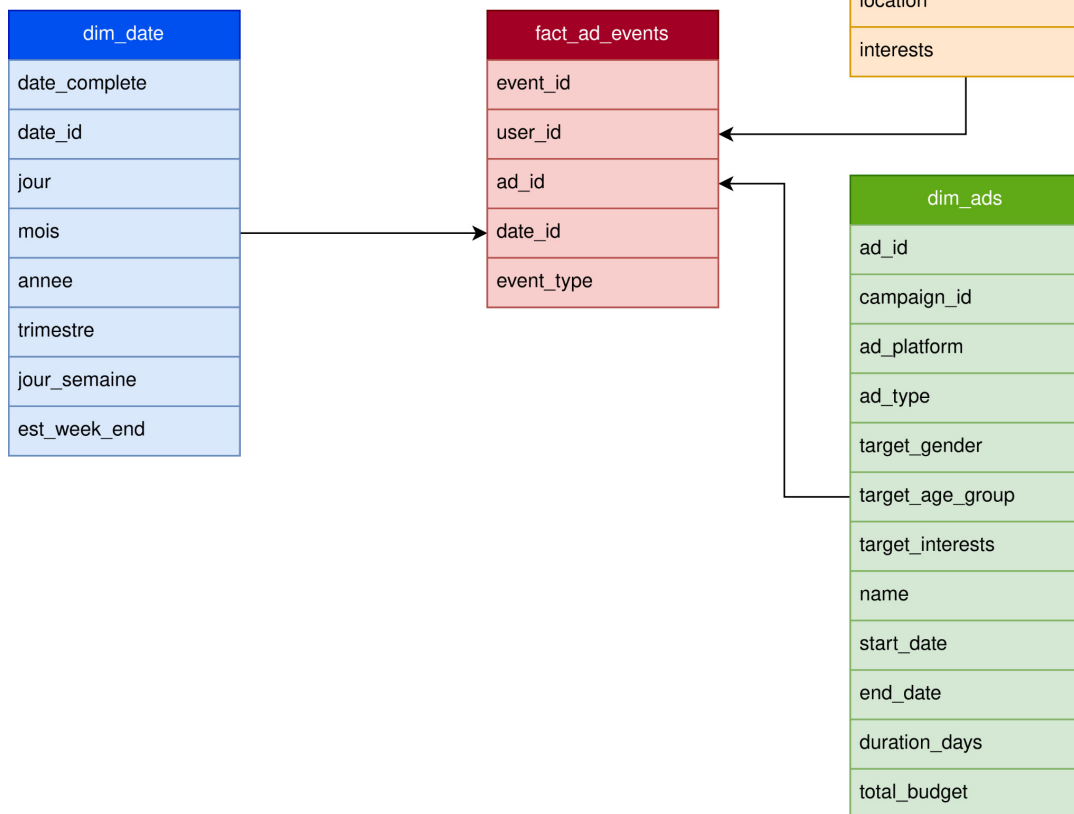


4. Description of the DW star schema mode

To optimize analytical performance and simplify report generation, the operational data was transformed into a Star Schema dimensional model (Dimensional/Silver layer).

- **fact_ad_events:** The core fact table of the model. It records granular ad interactions by storing foreign keys (`user_id`, `ad_id`, `date_id`) and the operational dimension `event_type`. It serves as the central bridge for all multi-dimensional analysis.
- **dim_users:** A conformed dimension table directly maps user profile specifics like age groupings and geographic attributes to allow seamless demographic slicing.
- **dim_ads:** This table denormalizes the original operational ads and campaigns tables into a single dimension. It collapses the relationship to provide analysts with immediate access to campaign timelines, budgets, and targeting strategies without traversing multiple tables.
- **dim_date:** A newly generated time dimension created during the ETL process from the operational timestamp. It translates raw time into comprehensive temporal attributes.

By establishing clear one-to-many relationships from the dimensions to the central fact table, this model ensures fast query execution and an intuitive data structure for BI tools.



5. Description of the ETL process

The Extract, Transform, Load (ETL) process serves as the data pipeline that bridges the operational database and the analytical data warehouse. Its primary objective is to clean, restructure, and transition the raw, normalized transactional data into an optimized star schema that facilitates rapid and intuitive business intelligence reporting.

5.1 Extract Step

Unlike traditional enterprise data warehouses that pull from a myriad of fragmented sources (such as APIs, CRMs, or external flat files), this architecture relies on a single, centralized source: a SQLite database.

Because the source is local and stable, the extraction phase is highly straightforward. Using Python and the pandas library, the pipeline establishes a direct connection to the database and executes comprehensive `SELECT * SQL` queries across the four operational tables (`ad_events`, `users`, `ads`, and `campaigns`). This process ingests the raw transactional records entirely into memory as Pandas DataFrames, ensuring that subsequent transformations do not lock or disrupt the operational source database.

5.2 Transform Step

The transformation phase is where the structural evolution from a normalized operational schema to an analytical star schema takes place. It is divided into three logical modeling actions:

- **Flattening Dimensions** (`dim_ads`): To eliminate complex multi-table joins for the BI layer, a left merge is performed between the `ads` and `campaigns` DataFrames using the common `campaign_id`. This collapses two operational concepts into a single, comprehensive advertising dimension.
- **Generating the Temporal Dimension** (`dim_date`): The operational timestamp in the events data is normalized into discrete dates. The pipeline then dynamically calculates rich chronological attributes required for analysis.
- **Structuring the Fact Table** (`fact_ad_events`): The central interaction table is streamlined. The original raw timestamp is replaced with the newly generated `date_id` foreign key. Non-essential attributes are stripped

away, leaving a lean fact table composed strictly of transactional foreign keys (`user_id`, `ad_id`, `date_id`) and the operational metric (`event_type`).

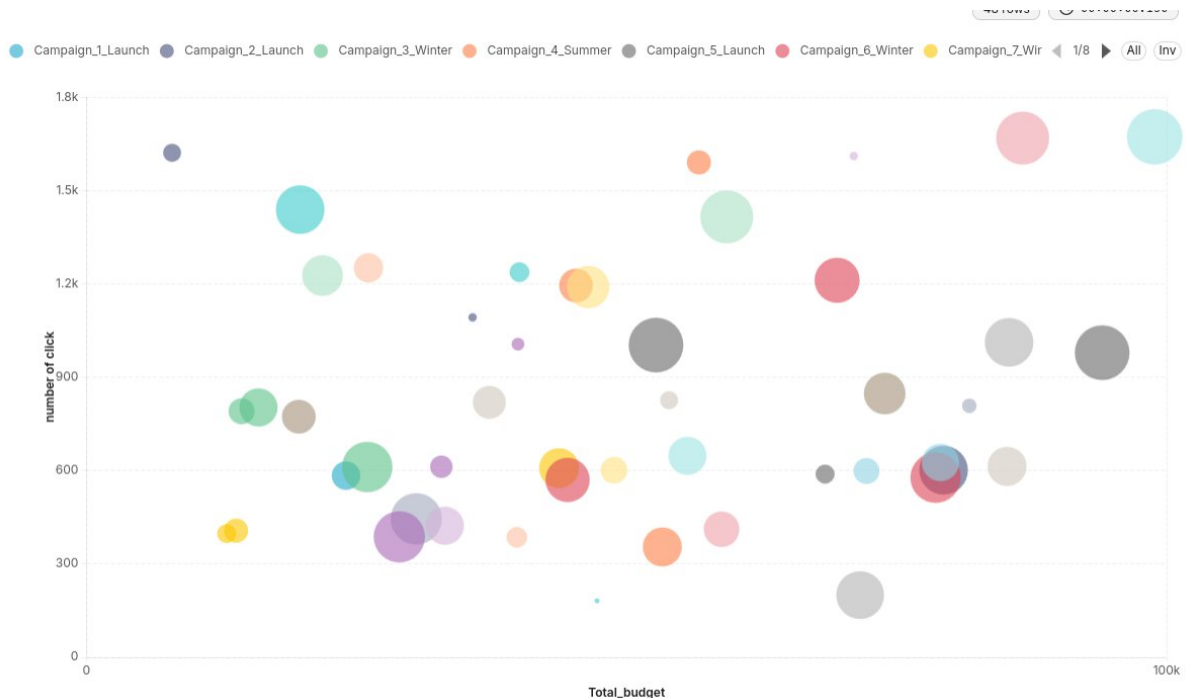
5.3 Load Step

The final phase involves persisting the transformed star schema data structures into the target destination environment, designated as the analytics data warehouse.

The pipeline initializes a connection to a new, separate SQLite instance dedicated exclusively to analytical workloads and BI tool connectivity. The four finalized DataFrames (`fact_ad_events`, `dim_users`, `dim_ads`, and `dim_date`) are written to this database.

6. Reports built in selected BI tool

6.1 Campaign Distribution by Number of Clicks and Total Cost



This bubble chart visualizes the Return on Investment (ROI) across marketing campaigns by mapping their total budget against the total number of user clicks generated. The size of each bubble corresponds to the campaign's operational duration in days, while different colors represent specific campaign names, allowing for a multi-dimensional performance comparison.

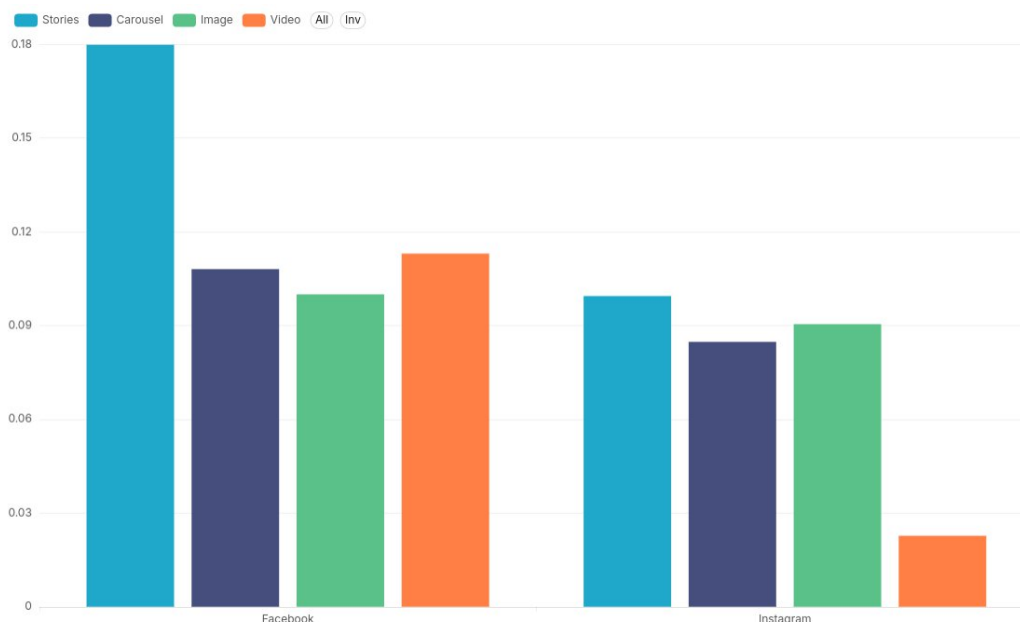
Key Insights & Results Analysis

- **Outstanding High-ROI Champions (Top-Left):** Several campaigns with low-to-mid budgets (between 15k and 30k) managed to maximize user engagement, scoring remarkably high between 1.4k and 1.6k clicks. These represent our most cost-effective and highly optimized assets.
- **Severe Budget Bleeding (Bottom-Right):** There is a critical performance anomaly on the higher end of the spending scale. Certain high-budget campaigns (spending between 70k and 90k) underperformed drastically,

pulling in fewer than 600 clicks—with one extreme outlier dropping near 200 clicks. This indicates a massive waste of ad spend.

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- **Lack of Budget-to-Performance Correlation:** The wide scatter of data points proves that increasing budget does not automatically scale traffic. Campaigns with a 20k budget are frequently matching or even outperforming those running at the maximum 100k budget limit, highlighting that targeting and creative format matter more than the amount spent.

6.2 Return on Investment (ROI) by Platform and Ad Type



This grouped bar chart evaluates marketing performance by breaking down the average campaign ROI (calculated as Total Clicks / Total Budget) across different advertising platforms (Facebook and Instagram). The data is further segmented by ad format—Stories, Carousel, Image, and Video—allowing for a direct comparison of asset efficiency between the two channels.

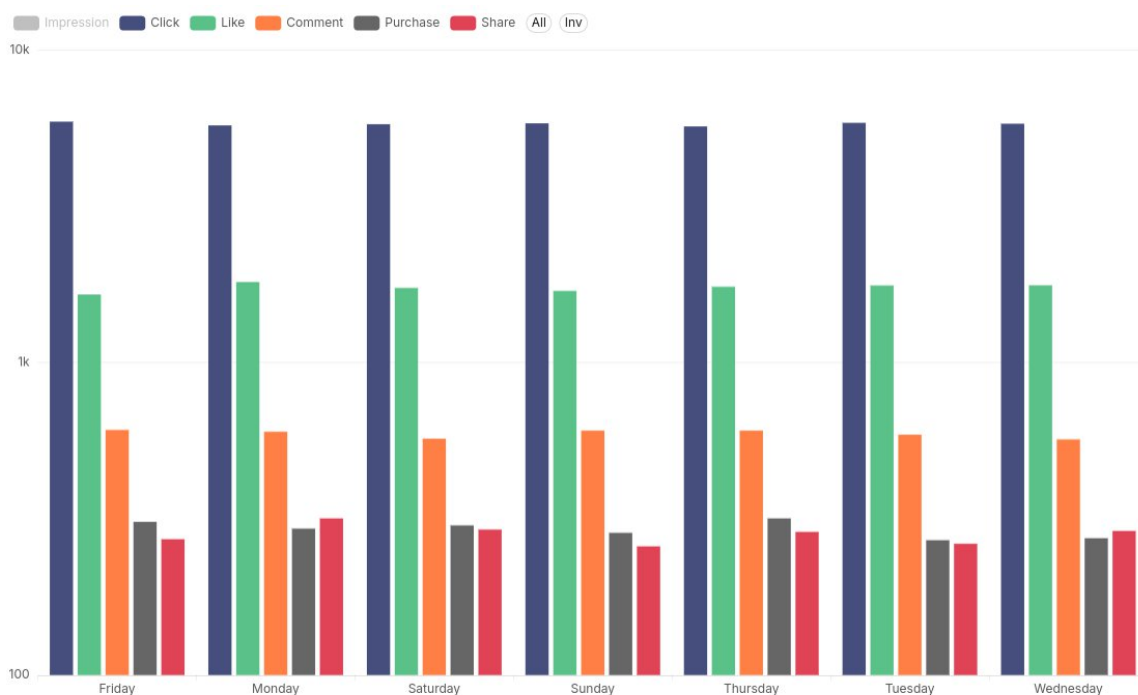
Key Insights & Results Analysis

- **Facebook Stories dominate efficiency (Top Performer):** The single highest performing asset type across the entire dataset is Facebook Stories, reaching an outstanding ROI peak of \$0.18\$. It generates

significantly more user engagement per dollar spent than any other placement.

- **The Video Format Polarisation:** There is a stark, fascinating contrast in how video content performs between the two channels. While Video is highly effective on Facebook (ranking as its second-best format at around \$0.11\$ ROI), it completely tanks on Instagram, dropping to the absolute lowest spot on the chart at less than \$0.03\$ ROI. This suggests our video creatives are heavily unoptimized for Instagram's feed/reels algorithm or audience.
- **Facebook Outperforms Instagram Across the Board:** When comparing identical formats, Facebook systematically yields a higher ROI than Instagram. Stories yield nearly double the return on Facebook (\$0.18\$ vs \$0.10\$), and Carousel shows a clear edge (\$0.11\$ vs \$0.085\$). The only format where Instagram holds its ground competitively is Static Images, keeping a steady ROI of \$0.09\$ on both platforms.
- **Strategic Recommendation:** I advice shift a portion of the Instagram Video budget towards Facebook Stories and Facebook Video assets to maximize total acquisition volume without increasing global ad spend.

6.3 Impressions by Day



This grouped bar chart illustrates the volume of user interactions across the different days of the week, broken down by event type: Click, Like, Comment, Purchase, and Share. To effectively display and compare high-volume metrics (such as clicks) alongside lower-volume conversion data (such as purchases) on a single view, the Y-axis utilizes a logarithmic scale ranging from 100 to 10,000 events.

Key Insights & Results Analysis

- **Remarkable Stability Throughout the Week:** The most striking takeaway is the extreme uniformity of the data. Regardless of the day—whether it is a weekday or the weekend—the volume for each engagement type remains almost perfectly flat.
- **The Conversion Funnel Breakdown:**
 - * **Top Funnel (High Volume):** Clicks consistently dominate the charts as the primary user action, holding strong at the top tier near the 5,000 to 6,000 mark every single day.
 - **Mid Funnel (Social Engagement):** Post interactions sit comfortably in the middle, with Likes hovering around 2,000 events, followed by Comments at approximately 500.
 - **Bottom Funnel (Conversions):** Purchases and Shares represent the tightest bottleneck, clustering at the lowest tier between 200 and 250 occurrences per day.
- **Logarithmic Interpretation Note:** Because the axis is logarithmic, the visual gap between bars minimizes massive numeric differences. For instance, while the "Click" bar looks only about three times taller than the "Purchase" bar, it actually represents roughly 25 to 30 times more events in reality.
- **Strategic Takeaway:** Since user behavior and purchase intent do not fluctuate based on the day of the week, there is no need to optimize or restrict ad delivery schedules (dayparting). A uniform budget distribution across the entire week remains the most logical approach for these campaigns.